manure allocation

Purpose

`manure_allo.mnu` sets up manure allocation: manure sources with a monthly production, and HRUs that receive manure from them when a decision table in `lum.dtl` asks for it. The file name is fixed. If the file is missing, manure allocation is off.

SWAT+ 62.0.1 reads this file and checks the decision tables every day, but applies no manure to any HRU. To apply manure, use a `fertilize` action in `lum.dtl` or a `manu` operation in `management.sch`.

Format

Free format, whitespace-separated.

- Line 1: title (skipped).
- Line 2: number of allocation objects.
- Then, for each allocation object:
  1. header line (skipped);
  2. object line: `name`, `rule_typ`, `src_obs`, `trn_obs`;
  3. header line (skipped);
  4. `src_obs` source lines;
  5. header line (skipped);
  6. `trn_obs` demand lines.

Object line

#	Column	Description
1	`name`	Allocation object name.
2	`rule_typ`	Allocation rule. Read but not used.
3	`src_obs`	Number of source lines.
4	`trn_obs`	Number of demand lines.

Source lines

#	Column	Units	Description
1	`num`	-	Source number, 1 to `src_obs`. The line is stored at this position.
2	`mois_typ`	-	Moisture class, `wet` or `dry`.
3	`manure_typ`	-	Manure name in `fertilizer.frt`.
4	`lat`	degrees	Latitude of the source.
5	`long`	degrees	Longitude of the source.
6	`stor_init`	t	Manure in storage at the start of the simulation.
7	`stor_max`	t	Storage capacity.
8 to 19	`prod_jan` … `prod_dec`	t/month	Manure produced in each month, January to December.

Demand lines

#	Column	Description
1	`num`	Demand number, 1 to `trn_obs`. The line is stored at this position.
2	`ob_typ`	Receiving object type: `hru`, `muni` (treatment) or `divert` (diversion out of the basin). Only `hru` demands are linked to a decision table.
3	`ob_num`	Number of the receiving object.
4	`dtbl`	Name of the decision table in `lum.dtl` that requests manure for this HRU, or `null`.
5	`right`	Manure right, `sr` (senior) or `jr` (junior).

The decision table requests manure with a `manure_demand` action: `obj_num` is the source number, `option` the application method in `chem_app.ops`, `const` the amount in t/ha and `const2` the maximum number of requests per year (see `*.dtl`).

Example

One allocation object with one source and one HRU demand:

```
manure_allo.mnu: example
1
name            rule_typ    src_obs  trn_obs
farm_a          fixed             1        1
num  mois_typ  manure_typ  lat       long      stor_init  stor_max   prod_jan  feb  mar  apr  may  jun  ju
  1  wet       beef_fr     41.99   -93.62          0.0       500.0    5.0  5.0  6.0  6.0  7.0  7.0  7.0  7
num  ob_typ    ob_num  dtbl                right
  1  hru            1  apply_manure         sr
```

With a table `apply_manure` in `lum.dtl` whose first action is `manure_demand` with source 1, method `broadcast` and 2 t/ha, the intent is to apply 2 t/ha of `beef_fr` from the source whenever the table's conditions are met. In 62.0.1 the request is recorded but nothing is applied.

Related files

- `fertilizer.frt`: the manure named in `manure_typ`.
- `*.dtl`: the decision tables named on demand lines and the `manure_demand` action.
- `chem_app.ops`: application methods.

Sources

- Reader: `manure_allocation_read.f90`, called from `main.f90.in` after `dtbl_lum_read`
- Types: `manure_allocation_module.f90` (`manure_allocation`, `manure_source_objects`, `manure_demand_objects`)
- Daily calculation: `mallo_control.f90`, called from `time_control.f90`
- `manure_demand` action: `actions.f90`, cross-walk in `dtbl_lum_read.f90`
- HRU decision table slot: `hru_dtbl_actions_init.f90`
- Output: `manure_source_output.f90`, `manure_demand_output.f90`, called from `command.f90`

Column	Fortran variable
object line	`mallo%name`, `%rule_typ`, `%src_obs`, `%trn_obs`
source `manure_typ`	`mallo%src%manure_typ` (index in `mallo%src%fertdb`)
source `prod_jan` … `prod_dec`	`mallo%src%prod_mon(1:12)`
demand `dtbl`	`mallo%trn%dtbl` (index in `mallo%trn%dtbl_num` and `hru%man_trn_dtbl`)
`manure_demand` `obj_num`, `const`, `option`	`manure_amt%src_obj`, `%app_t_ha`, `%app_method`